

ALEXANDROUPOLI, Greece

WMO: 166270

Lat: 40.856N

Lon: 25.956E

Elev: 7

StdP: 101.24

Time Zone: 2.00 (EUE)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.7	(c) -2.9	(d) -9.8	(e) 1.6	(f) -2.8	(g) -7.8	(h) 1.9	(i) -0.5	(j) 15.2	(k) 1.1	(l) 13.8	(m) 6.1	(n) 3.6	(o) 20	(p) 0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	34.8	21.6	33.1	21.3	31.9	20.8	23.6	30.6	22.7	30.0	22.0	29.3	5.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	15.8	27.7	20.2	14.9	26.9	19.2	14.0	25.9	70.5	30.8	67.1	29.9	64.3	29.2	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.9	(b) 9.4	(c) 8.4	(e) -7.8	(f) 36.7	(g) 2.1	(h) 1.8	(i) -9.3	(j) 38.0	(k) -10.5	(l) 39.1	(m) -11.7	(n) 40.1	(o) -13.2	(p) 41.4		
(4) 10.9	(5) 9.4	(6) 8.4	(7) -8.7	(8) 25.2	(9) 2.0	(10) 0.9	(11) -10.1	(12) 25.9	(13) -11.3	(14) 26.4	(15) -12.4	(16) 26.9	(17) -13.9	(18) 27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	5.2	6.1	8.6	12.7	18.0	22.6	25.5	25.6	20.9	15.8	10.9	6.8		
	DBStd	7.99	4.02	3.91	3.46	3.05	2.76	2.89	2.23	2.40	2.81	3.32	3.90	4.65		
	HDD10.0	513	156	118	69	11	0	0	0	0	0	2	35	122		
	HDD18.3	1946	407	343	302	169	39	2	0	0	9	91	225	358		
	CDD10.0	2315	7	8	25	92	249	378	481	483	327	182	61	22		
	CDD18.3	707	0	0	0	1	30	130	222	225	87	12	1	0		
	CDH23.3	7154	0	0	0	9	243	1273	2474	2444	661	51	0	0		
	CDH26.7	2812	0	0	0	1	36	439	1079	1098	154	4	0	0		
(14) Wind	WSAvg	3.3	3.5	3.5	3.6	3.0	2.9	2.8	3.3	3.3	3.1	3.4	3.1	3.7		
(15) Precipitation	PrecAvg	554	63	60	51	39	36	29	17	10	31	49	74	90		
	PrecMax	796	183	186	157	84	108	110	67	68	153	158	231	254		
	PrecMin	336	6	2	2	6	8	0	0	0	0	0	8	0		
	PrecStd	114	46	43	39	22	25	28	15	14	38	42	49	59		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	16.2	19.2	24.6	29.8	35.2	36.8	37.2	32.2	27.0	22.0	19.2		
		MCWB	13.8	12.9	13.2	16.3	18.0	22.0	21.9	22.1	19.9	19.1	17.5	16.6		
	2%	DB	14.9	15.0	17.1	21.9	27.2	32.8	34.3	35.0	30.0	24.9	20.0	16.9		
		MCWB	12.7	12.2	12.9	15.0	18.1	21.4	21.2	21.8	19.7	18.2	16.7	14.7		
	5%	DB	13.4	13.9	16.0	20.1	26.0	31.0	33.1	33.2	28.2	23.1	18.2	15.2		
		MCWB	11.2	11.5	12.5	14.2	17.9	20.5	21.2	21.4	19.4	17.6	15.3	13.6		
	10%	DB	11.9	12.8	14.8	18.8	24.2	29.1	31.9	32.0	27.0	21.8	17.1	14.0		
		MCWB	10.0	10.4	11.7	13.6	17.4	19.9	20.8	20.9	19.1	17.0	14.5	12.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	13.5	14.8	17.2	20.9	24.1	24.6	24.7	23.1	21.1	18.2	16.8		
		MCDB	16.0	15.6	17.7	22.4	25.8	30.6	32.3	31.5	27.4	24.2	20.6	18.9		
	2%	WB	12.9	12.6	13.7	15.9	19.7	22.6	23.4	23.7	21.9	19.7	17.0	15.0		
		MCDB	14.6	14.5	16.3	20.0	24.7	29.7	30.8	30.9	26.4	22.8	19.4	16.5		
	5%	WB	11.5	11.6	12.8	15.1	18.7	21.7	22.5	22.7	20.9	18.6	15.9	13.9		
		MCDB	13.2	13.6	15.3	19.0	23.9	28.8	30.2	30.5	25.9	21.8	18.0	15.4		
	10%	WB	10.1	10.5	11.9	14.3	17.9	20.9	21.7	22.0	19.9	17.4	14.9	12.5		
		MCDB	11.5	12.3	14.4	17.9	23.4	27.7	29.5	29.8	25.7	20.9	16.9	14.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.9	8.6	9.8	10.9	11.2	11.7	11.8	10.8	9.1	8.3	6.8		
		MCDBR	7.8	8.9	9.6	11.8	13.1	13.0	13.3	13.6	12.5	10.6	8.9	6.3		
	5% WB	MCWBR	6.4	6.9	6.6	7.0	6.8	5.9	5.8	5.7	6.0	6.2	6.5	5.4		
		MCWBR	7.2	8.1	8.7	10.3	11.5	11.8	12.0	12.3	11.0	9.2	8.2	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.364	0.396	0.440	0.485	0.458	0.452	0.450	0.456	0.443	0.417	0.389	0.373		
	taud		2.330	2.219	2.107	2.003	2.159	2.220	2.230	2.204	2.212	2.255	2.279	2.289		
	Ebn at Noon		771	798	802	792	825	830	827	810	791	764	731	721		
	Edh at Noon		89	115	143	170	149	140	138	138	128	110	93	87		
(44) All-Sky Solar Radiation	RadAvg		1.68	2.52	3.76	5.11	6.47	7.12	7.41	6.58	4.89	3.18	2.03	1.49		
	RadStd		0.21	0.24	0.34	0.53	0.39	0.49	0.32	0.23	0.32	0.30	0.20	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-182	+206	+125
(47) Regional (14 neighbors)		+0.51	N/A	N/A	N/A	+0.49	+0.46	-64	-144	+116	+59

Nomenclature: See separate page